

Effects of top-down and bottom-up attention on SPCN & RLpcN

(Rashal et al., 2023)

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Background



Background

Definitions

The **N2pc** is a negative ERP component that typically appears around 200-300ms after stimulation. It appears in the parietal and occipital areas of the brain, always contralateral to the visual stimulus



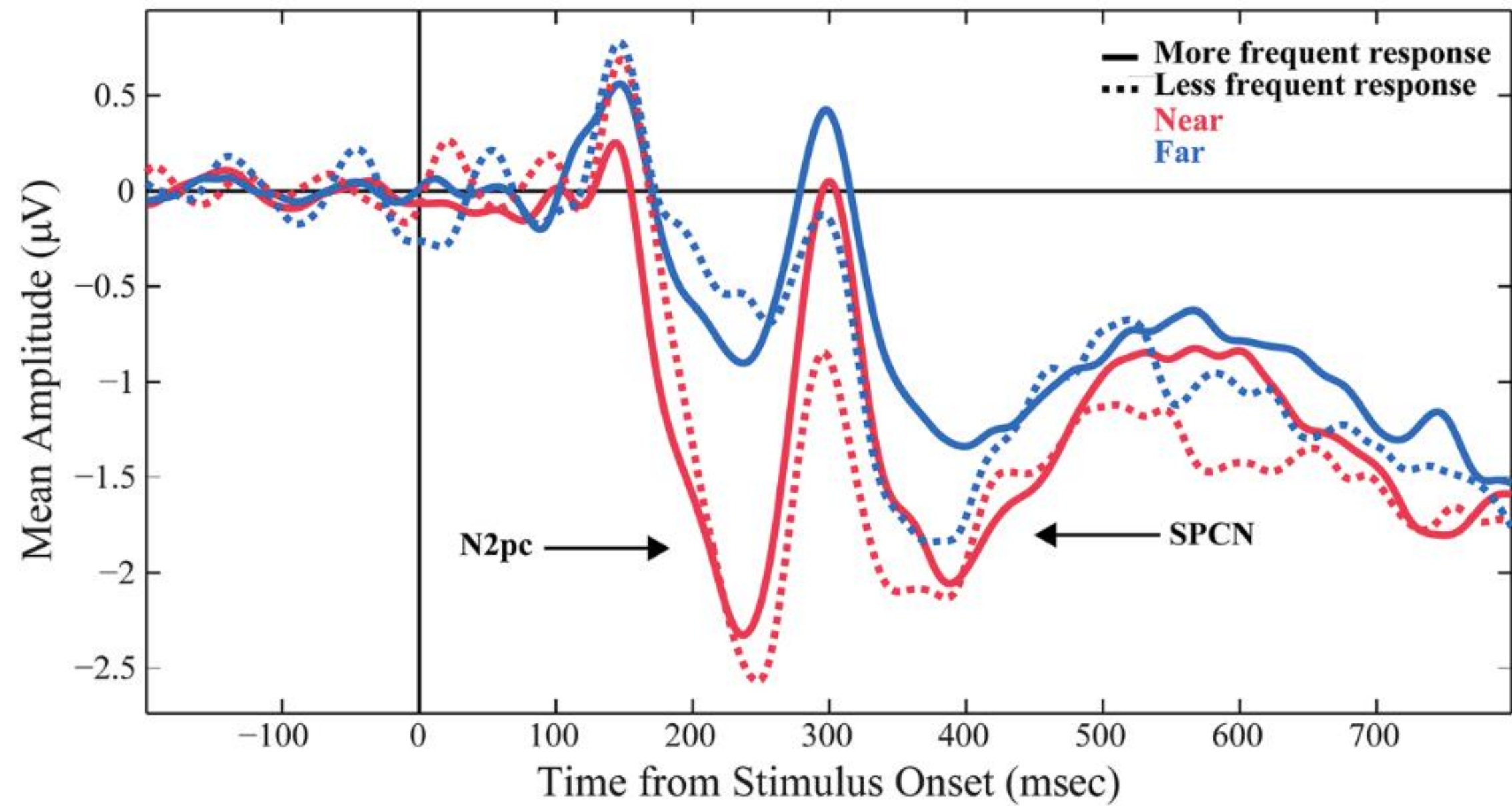
Background

Definitions

The sustained posterior contralateral negativity (**SPCN**) is a negative deflection observed in the parietal and occipital areas

It appears at around **400ms** in the hemisphere contralateral to the stimulus

Notably, its amplitude is positively correlated with the number of visual stimuli (Chen et al. 2022)



Drisdelle & Jolicoeur (2019)

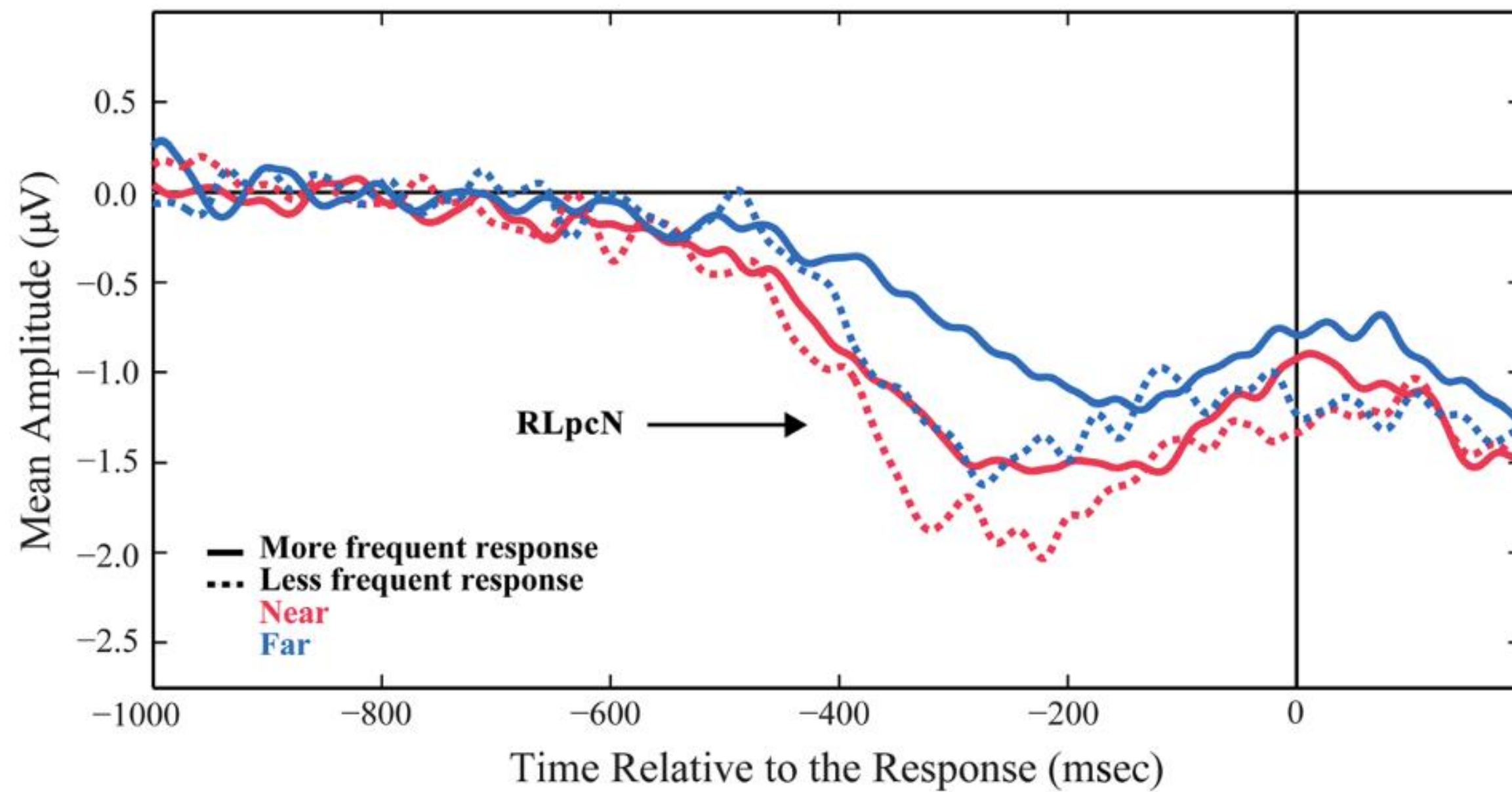
Background

Definitions

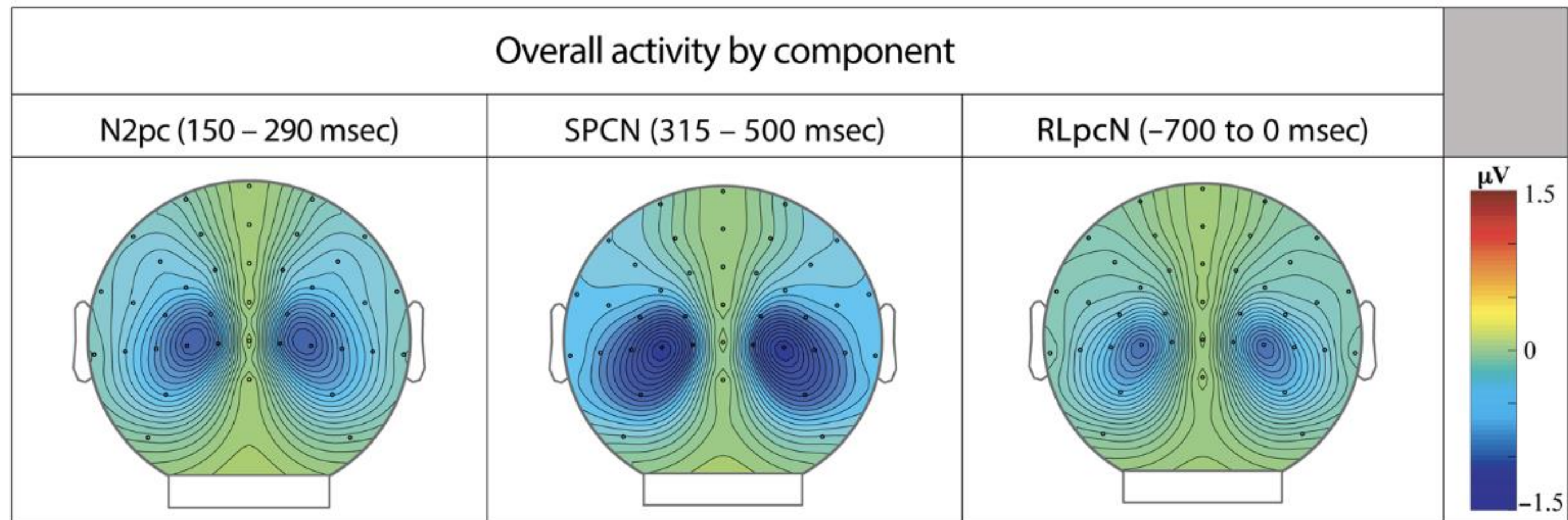
The response-locked posterior contralateral negativity (**RLpcN**) has been recently studied in relation to visual attention

It appears around **-400ms** as the participant prepares their response to visual stimuli

RLpcN duration is directly correlated with **response times**, and often occurs concurrently with the SPCN



Drisdelle & Jolicoeur (2019)



Drisdelle & Jolicoeur (2019)



Background

Literature

Drisdelle & Jolicoeur (2019):

This study measured the N2pc, SPCN, and RLpcN. The results showed that RLpcN variance is correlated with attention, processing, and response time



Background

Literature

Rashal et al. (2022):

The previous iteration of Rashal et al.'s experiment measured the **N2pc** and contingent negative variation (**CNV**) instead of the **SPCN** and **RLpcN**

Effects of cues and salient distractors were shown

Rashal et al. hypothesized that attention is affected by top-down cueing

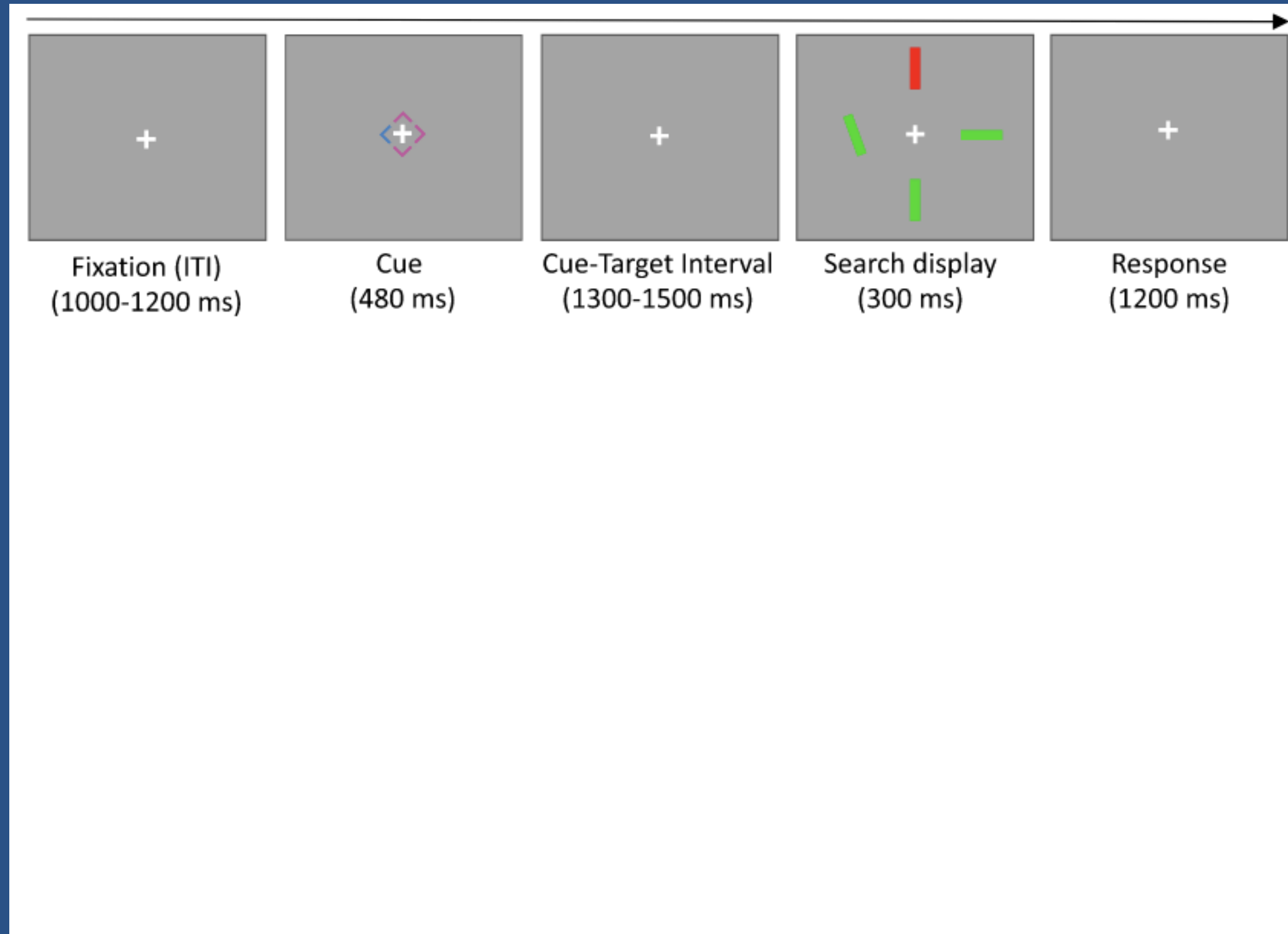


Table 1 Accuracy (%) and reaction times (ms) in the different cue and display conditions in Experiments 1 and 2 reported in Rashal et al. (2022)

		Experiment 1			
		Accuracy (%)		Reaction Time (ms)	
Cue	Display	Mean	<i>SD</i>	Mean	<i>SD</i>
Neutral	No distractor	91.0	7.2	612	115
	Salient target	92.8	5.8	603	111
	Lateral target	91.3	6.4	638	121
	Midline target	92.3	7.5	688	129
Valid	No distractor	95.1	3.6	544	109
	Salient target	94.1	4.3	541	102
	Lateral target	94.9	3.9	557	115
	Midline target	97.1	2.7	561	108

Rashal et al. (2022)

Hypotheses

Hypotheses

The **RLpcN** has two segments:

RLpcN1 is correlated with search difficulty

RLpcN2 is correlated with response preparation and selection

RLpcN1 should be more negative in **Experiment 2**, as should the **SPCN**

Hypotheses

In visual search tasks, both the **SPCN** and **RLpcN** should be affected by task difficulty and target salience

There should be an interaction effect between target salience and task difficulty on **SPCN** and **RLpcN2**, and this effect should be larger in **Experiment 2**

Methods



Methods

Participants

38 volunteers:

21 in Experiment 1 (mean = 22.44)

17 in Experiment 2 (mean = 23.43)

Between-subjects design

1,152 total trials: 10 trials of practice and 12

blocks of 96 experimental trials



Methods

General setup

Stimuli (4 bars, either green or red, 3 of the same color and 1 outlier) appeared at various orientations on a grey background



Methods

Conditions

Salient-target condition: target bar is a different color than the three distractors

No-distractor condition: all bars are the same color

Lateral-target condition: one distractor bar is a different color and appeared on the lateral midline, and the target (the same color as the other two bars) also appeared on the lateral midline



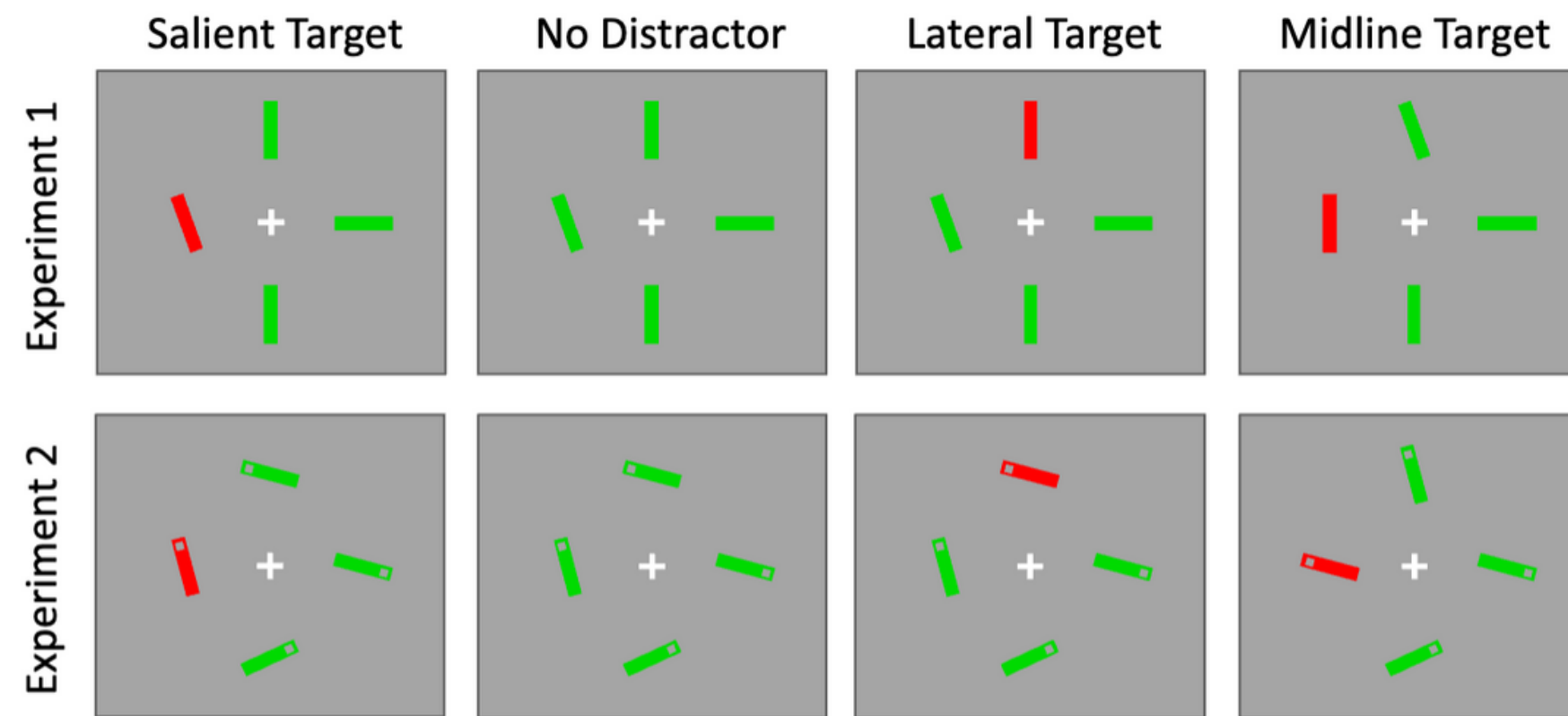
Methods

Cues

Before stimuli appeared, a diamond would briefly be presented around the fixation cross

Neutral cue: magenta diamond

Valid cue: magenta diamonds with one cyan corner

A



Methods

EEG analysis

10-10 system

Electrodes of interest: PO7, PO8

High pass of 0.1 Hz

ICA analysis to remove artifacts

Semi-automatic artifact rejection



Methods

EEG analysis

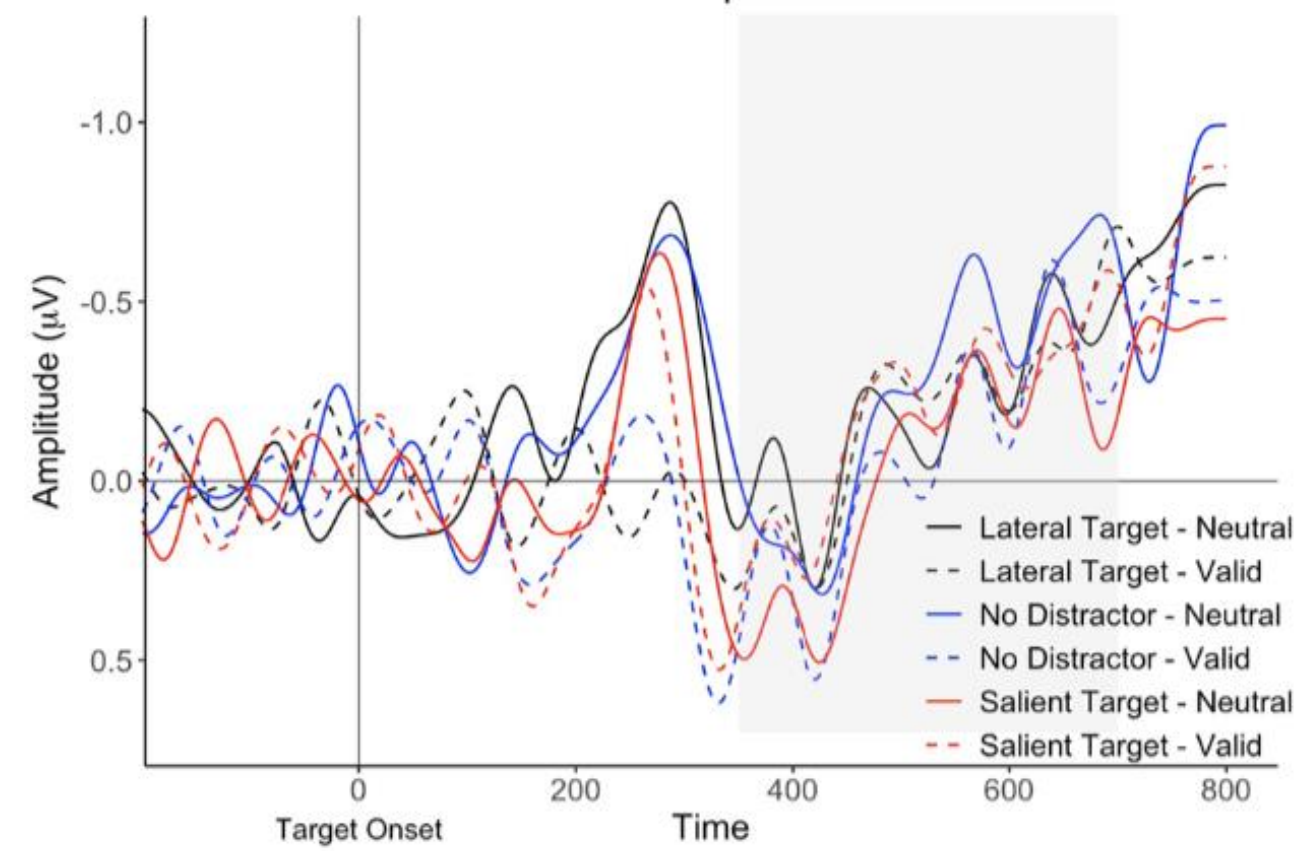
Trials rejected: 6% in Experiment 1 and 11% in Experiment 2

SPCN analysis: epochs from -200 to 800ms, measured from 350 to 700ms after stimulus presentation

RLpcN analysis: epochs from -1000 to 2000ms, measured in two segments of 300ms each

Results

SPCN in Experiment 1





Results

SPCN: Mixed ANOVA

Within-subjects factors: search display (lateral-target, no-distractor, salient-target) x cue (valid, neutral)

Between-subjects factor: experiment (1, 2)

Significant main effect of experiment

($p = 0.01$)



Results

SPCN: Mixed ANOVA

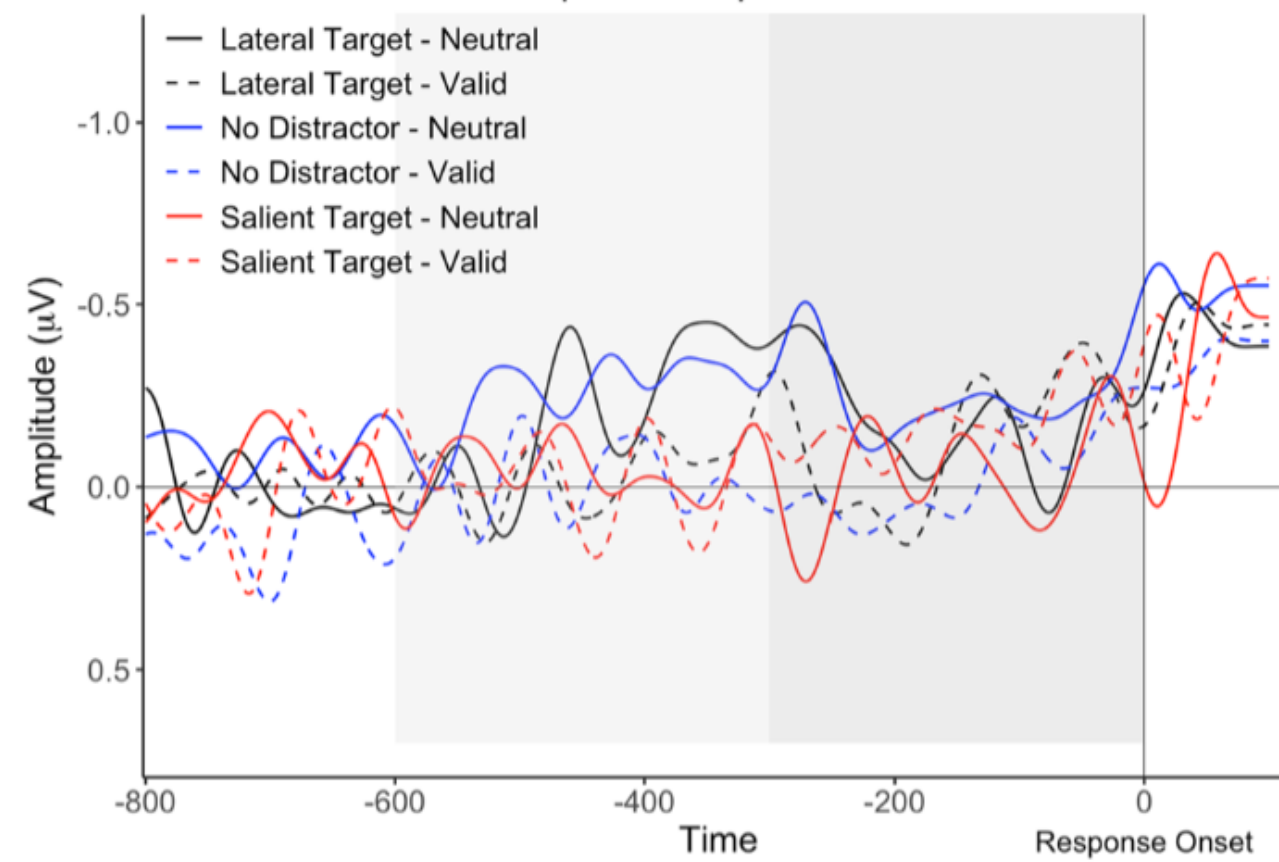
Within-subjects factor: search display (no-distractor, salient-target)

Between-subjects factor: experiment (1, 2)

Significant main effect of experiment

($p = 0.005$)

RLpcN in Experiment 1





Results

RLpcN: Mixed ANOVA

Within-subjects factors: segment (first, second)
x search display (lateral-target, no-distractor,
salient-target) x cue (valid, neutral)

Between-subjects factor: experiment (1, 2)

Significant main effects of experiment ($p = 0.006$) **and segment** ($p < 0.0001$)

Interaction effects between all factors



Results

RLpcN: Mixed ANOVA

Within-subjects factor: search display (no-distractor, salient-target)

Between-subjects factor: experiment (1, 2)

Main effects of experiment ($p = 0.0009$) **and search display** ($p = 0.03$)

Interaction effect between experiment and search display ($p = 0.006$)



Results

RLpcN: rm-ANOVA

Search display x segment

Experiment 1:

No significant effects

RLpcN2 was more negative than RLpcN1

Experiment 2:

Main effect of segment ($p = 0.0001$)

Interaction effect between segment and search display ($p < 0.0001$)

Main Findings



Findings

SPCN

Minimal lateralization seen in certain conditions in Experiment 1, but **significant lateralization was seen in all conditions in Experiment 2**, confirming that SPCN is affected by task difficulty

SPCN area was larger for salient-target condition in Experiment 2



Findings

RLpcN1

Significant lateralization seen in certain conditions in Experiment 1 & 2



Findings

RLpcN2

Significant lateralization seen in certain conditions in Experiment 1

Significant lateralization seen in all conditions following a valid cue in Experiment 2

Area was larger in Experiment 2

Discussion



Discussion

Recap

Aim was to study visual working memory, post-selection processes, and top-down/bottom-up attention by analyzing **SPCN** and **RLpcN**



Discussion

Results

Experiment 2, which was more difficult and complex, had a strong effect on **SPCN** and **RLpcN2**

RLpcN1 was affected by **salient distractors** (differently colored distractor bars) and

RLpcN2 was affected by **target salience** (differently colored target bar)



Discussion

Results

Salient distractors had no effect on SPCN or RLpcN2

Search display did not have an effect on RLpcN in Experiment 1



Discussion

Results

Cues had little to no effect on RLpcN or SPCN. In Rashal et al. (2022), an effect on the **N2pc** was seen, probably due to the timing of the experiment

What's next?



Sources

Chen, Y., Brigadoi, S., Schiano Lomoriello, A., Jolicoeur, P., Simal, A., Fu, S., Baro, V., & Dell'Acqua, R. (2022). A bilateral SPCN is elicited by to-be-memorized visual stimuli displayed along the vertical midline. *Psychophysiology*, 59(8), e14045. [<https://doi.org/10.1111/psyp.14045>]

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Rashal, E., Santandrea, E., Ben-Hamed, S., Macaluso, E., Chelazzi, L., & Boehler, C. N. (2023). Effects of top-down and bottom-up attention on post-selection posterior contralateral negativity. *Attention, Perception, & Psychophysics*, 85(3), 705–717. <https://doi.org/10.3758/s13414-022-02636-7>

Thank you for
your attention!

Questions?

